

Toran Bahadur Niroula | Aerospace Engineer

Lalitpur, Nepal

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🔬 Research Profile

Aerospace engineer with research experience in experimental aerodynamics, optical diagnostics, particle image velocimetry, computational fluid dynamics, and scientific computing. Experienced in PIV/POD-based flow analysis, quantitative Rainbow Schlieren Deflectometry, OpenFOAM, and development of Python-based research tools. Research interests include high-speed and unsteady flows, turbulence, experimental fluid mechanics, fluid-structure interaction, and data-driven flow analysis.

🎓 Education

Institute of Engineering, Pulchowk Campus, Tribhuvan University **Lalitpur, Nepal**
Bachelor of Engineering in Aerospace Engineering **2022–2026**

- **Cumulative Percentage:** 82.62% **Academic Standing:** Top 4 in graduating class
- **Relevant Coursework:** Fluid Dynamics, Compressible Aerodynamics, Computational Fluid Dynamics, Numerical Methods, Aircraft Structures, Flight Dynamics, Control Systems, UAV Design, Aircraft Preliminary Design

🔬 Research Experience

University of Birmingham **Remote**
Research Collaboration – BioFSI Lab **2026–present**

- Investigating the flight mechanics and vortex formation of **dandelion seeds** using computational fluid dynamics and overset-mesh methods.
- Exploring fluid-structure interaction approaches for modelling the flexible pappus structure and its influence on aerodynamic behaviour.

Indian Institute of Technology Kanpur **Kanpur, India**
Research Intern – Experimental Aerodynamics Group **Oct–Nov 2025**

- Processed and analyzed **500 PIV snapshots** of a NACA0015 airfoil operating in reverse flow at 12° angle of attack.
- Extracted mean velocity, vorticity, Q-criterion, turbulence intensity, and Reynolds-stress fields using PIVlab and Python/MATLAB workflows.
- Performed **Proper Orthogonal Decomposition (POD)** and modal analysis to identify dominant coherent structures in the separated wake.
- Gained experimental experience with tuft visualization and aerodynamic force/torque measurements in rotorcraft flow-control studies.

Indian Institute of Technology Bombay **Remote**
CFD Research Intern – FOSSEE Fellowship **Feb–Jun 2025**

- Migrated a published fixed-wing UAV aerodynamic study from ANSYS Fluent to **OpenFOAM** using simpleFoam, Spalart–Allmaras turbulence modelling, and snappyHexMesh.
- Performed mesh-independence and turbulence-model sensitivity studies and compared reproduced baseline results with the reference study.
- Extended the study to blended winglets at different cant angles, investigating their influence on lift-to-drag performance and wingtip vortex structures.

⚙️ Research Projects

2025–2026: Quantitative Rainbow Schlieren Deflectometry for Surface-Heating Effects in a Compressible Wall Jet

- Designed and constructed a **Rainbow Schlieren Deflectometry (RSD)** facility to quantitatively image density-gradient fields around a heated compressible wall jet over a temperature-controlled flat plate.
- Developed image-processing and quantitative reconstruction methods to convert raw RSD images into displacement, density-gradient, and density fields.
- Developed **pyrsd**, a Python research package for processing Rainbow Schlieren Deflectometry data.
- Compared experimental measurements with CFD and investigated a natural-convection case as an independent validation dataset.

2026: Hypersonics and Advanced Propulsion – Shock Interactions and Scramjet Intake

- Extended undergraduate hypersonics coursework through self-directed CFD studies using rhoCentralFoam.
- Investigated shock-shock interactions and compared numerical pressure distributions with local-inclination and shock-polar predictions.
- Designed and evaluated a single-ramp scramjet intake at Mach 6 using inviscid and RANS simulations.

Research Outputs

2026: Niroula, T. B. et al. *pyrsd: A Python Package for Processing Rainbow Schlieren Deflectometry Data*. Research software, Zenodo. doi:10.5281/zenodo.19219233

2026: *Study of the Surface Heating Effect on a Compressible Wall Jet Using Rainbow Schlieren Deflectometry*. Undergraduate Final Year Project, IOE Pulchowk Campus.

2025: *Flow Analysis and Modal Decomposition of a NACA0015 Airfoil Under Reverse Flow Configuration Using PIV Data*. Research internship report, Experimental Aerodynamics Group, IIT Kanpur.

2025: *Numerical Simulation of a Fixed-Wing UAV and the Effect of Blended Winglets*. Research internship report, FOSSEE Fellowship, IIT Bombay.

⚙️ Technical Skills

CFD & Simulation: OpenFOAM (rhoCentralFoam, simpleFoam, snappyHexMesh), ANSYS Fluent, ParaView

Experimental Methods: Rainbow Schlieren Deflectometry, Schlieren, Particle Image Velocimetry, Tuft Visualization

Flow Analysis: PIVlab, POD, Modal Analysis, Vorticity, Q-criterion, Reynolds-stress analysis

Scientific Computing: Python (NumPy, Matplotlib), MATLAB, LaTeX

CAD & Aircraft Design: CATIA, SolidWorks, XFLR5, X-Plane

✈️ Engineering Experience

AIAA Design/Build/Fly **IOE Pulchowk**
Design Team Member – DBF Pulchowk *2023–2024*

- Contributed to aerodynamic and stability design of the competition aircraft *Junkiri*, including aircraft sizing, stability analysis, flight testing, and technical documentation.

2025: Twin-Boom UAV Design and Fabrication

- Designed and fabricated a twin-boom fixed-wing UAV with a three-member team; performed XFLR5 aerodynamic analysis, hot-wire foam cutting, and flight testing.

2026: NIMBUS – Preliminary Design and Flight Dynamics of an Autonomous Agricultural UAV

- Contributed to conceptual sizing and configuration development of an electric agricultural UAV for a 200 kg payload and performed aerodynamic, performance, and flight-dynamics analysis.

👤 Teaching Experience

Department of Aerospace Engineering, IOE Pulchowk Campus **Lalitpur, Nepal**
Aerodynamics Practical Instructor *2025–2026*

Conducted practical/laboratory sessions in aerodynamics for third-year aerospace engineering students.

Department of Aerospace Engineering, IOE Pulchowk Campus **Lalitpur, Nepal**
Guest Lecturer – Instrumentation Systems *2025–2026*

Delivered guest lectures on instrumentation systems to second- and fourth-year aerospace engineering students.

🔍 Additional Research

2026–present: Wing–Propeller Aerodynamic Interaction

- Developing an OpenFOAM-based study of unsteady propeller–wing interaction, focusing on wake development and aerodynamic coupling.